

LAB 9: THREAD CREATION USING PTHREAD

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C PROGRAM

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <unistd.h>

void *thread_function(void *arg){
    int i;
    for(i = 1; i <= 5; i++){
        printf("Thread Executing: %d", i);
        sleep(1);
    }
    pthread_exit(NULL);
}

int main()
{
    pthread_t t1, t2;
    pthread_create(&t1, NULL, thread_function, NULL);
    pthread_create(&t2, NULL, thread_function, NULL);
    pthread_join(t1, NULL);
    pthread_join(t2, NULL);
    printf("All Threads Completed");
    return 0;
}
```

SHELL SCRIPTING

```
#!/bin/bash
task1()
{
    for i in 1 2 3 4 5
    do
        echo "Thread 1: $i"
        sleep 1
    done
}
task2()
{
    for i in 1 2 3 4 5
    do
        echo "Thread 2: $i"
        sleep 1
    done
}
task1 &
task2 &
wait
echo "All Threads Completed"
```

OUTPUT

```
(kali@Harish)-[~]  
└─$ nano threading.sh  
(kali@Harish)-[~]  
└─$ chmod +x threading.sh  
(kali@Harish)-[~]  
└─$ bash threading.sh  
Thread 1: 1  
Thread 2: 1  
Thread 1: 2  
Thread 2: 2  
Thread 1: 3  
Thread 2: 3  
Thread 1: 4  
Thread 2: 4  
Thread 1: 5  
Thread 2: 5  
All Threads Completed
```